MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL
PHYSICS

Motion

1.2

STUDENT NAME

0315 0805775

PHYSICS WITH
MOIZ DAWOODI

Motion

SYLLABUS OUTLINE:

1.2 Motion

1 Define speed as distance travelled per unit time and define velocity as change in displacement per unit time

2 Recall and use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$v = \frac{s}{t}$$

3 Recall and use the equation

$$\text{average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

4 Define acceleration as change in velocity per unit time; recall and use the equation

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a = \frac{\Delta v}{\Delta t}$$

5 State what is meant by, and describe examples of, uniform acceleration and non-uniform acceleration

6 Know that a deceleration is a negative acceleration and use this in calculations

7 Sketch, plot and interpret distance–time and speed–time graphs

8 Determine from the shape of a distance–time graph when an object is:
- (a) at rest
- (b) moving with constant speed
- (c) accelerating
- (d) decelerating

9 Determine from the shape of a speed–time graph when an object is:
- (a) at rest
- (b) moving with constant speed
- (c) moving with constant acceleration
- (d) moving with changing acceleration

10 State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately $9.8\,\text{m}/\text{s}^2$

11 Calculate speed from the gradient of a distance–time graph

12 Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration

13 Calculate acceleration from the gradient of a speed–time graph

DISTANCE.

"The actual path covered by a body is called distance."

Its a scalar quantity. Its S.I unit is meter (m).

DISPLACEMENT:

"Shortest distance from starting to ending point is called displacement."

It's a vector quantity. It's S.I unit is meter (m).

SPEED:

"The distance covered per unit time is called speed."

Its a scalar quantity. It's S.I unit is meter/second (m/s or ms^{-1}).

FORMULA:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\boxed{v = \frac{d}{t}}$$

AVERAGE SPEED:

"The total distance covered per total time is called average speed."

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$\boxed{\langle v \rangle = \frac{T.d}{T.t}}$$

VELOCITY:

"The displacement covered per unit time is called velocity."

Its a vector quantity. It's S.I unit is meter/second (m/s or ms^{-1}).

FORMULA:

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

$$\boxed{v = \frac{d}{t}}$$

UNIFORM VELOCITY:

"When body covers same displacement in every interval of time, the velocity is called uniform velocity."

ACCELERATION:

"Change in velocity per unit time is called acceleration."

Its a vector quantity. It's S.I unit is meter/second^2 (m/s^2 or ms^{-2}).

FORMULA:

$$\text{Acceleration} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time taken}}$$

$$\boxed{a = \frac{v - u}{t}}$$

TYPES OF ACCELERATION.

- **POSITIVE ACCELERATION:**

"An acceleration in which velocity of a body increases."

- **NEGATIVE ACCELERATION (DECCELERATION/RETARDATION):**

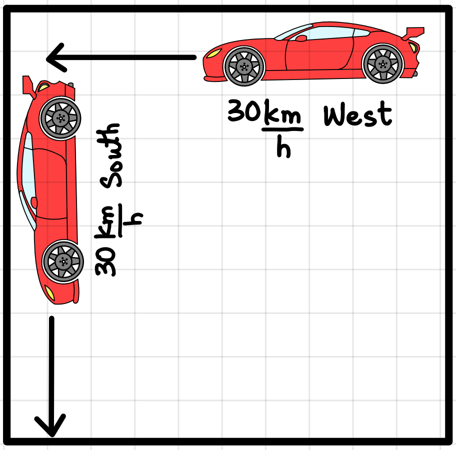
"An acceleration in which velocity of a body decreases."

- **ZERO ACCELERATION:**

"Acceleration is said to be zero if a body is at rest or moves with uniform velocity."

- **SPECIAL CASE**

A car moves with a velocity of 30 km/h towards west turns to the south with same magnitude. Will there be an acceleration?



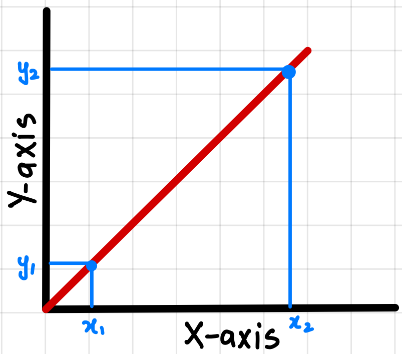
Yes, there will be an acceleration as acceleration depends upon change in velocity & velocity changes with direction.

> Velocity changes if there is a change in magnitude or change in direction.

UNIFORM ACCELERATION:

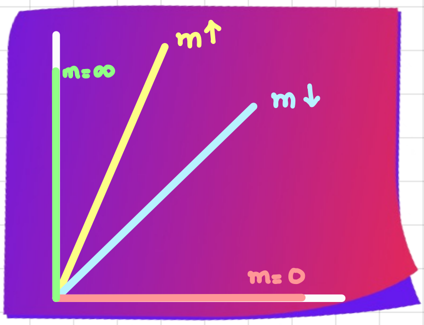
"When increased in velocity is same in every interval of time, the acceleration is called uniform acceleration."

MOTION GRAPHS:

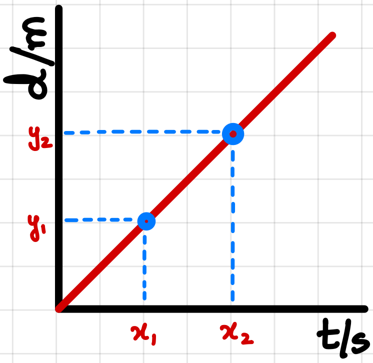


$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

> Straight line has uniform gradient

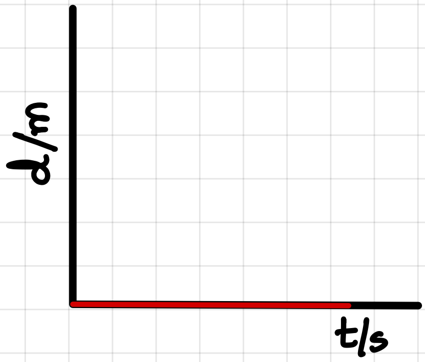


Displacement-Time Graph:

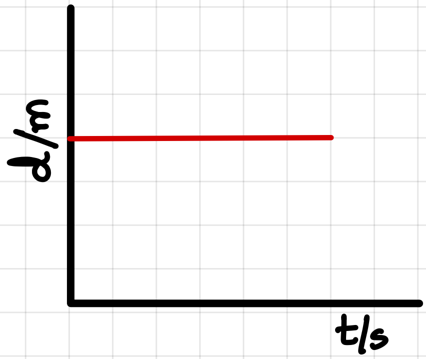


Δ Delta (Represents difference).

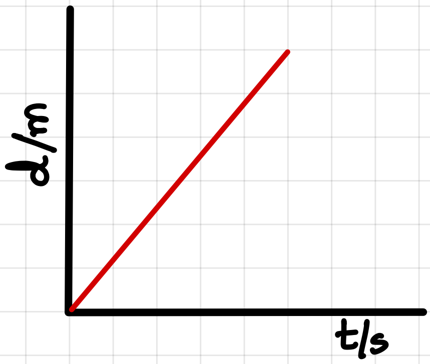
$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\Delta d}{\Delta t} = \text{velocity}$$



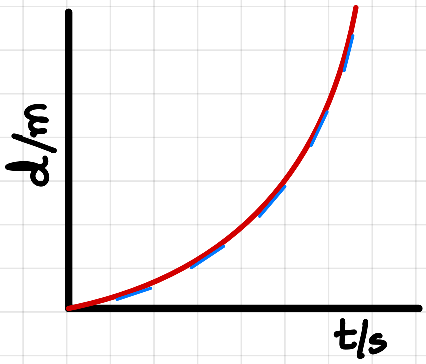
and



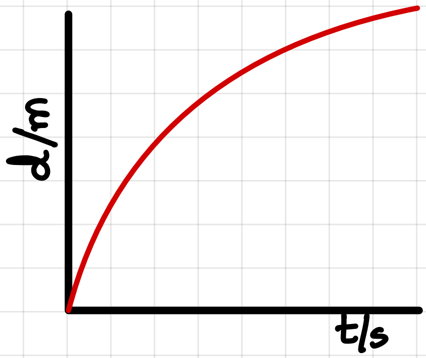
Body is at rest. v=0m/s



Uniform Velocity

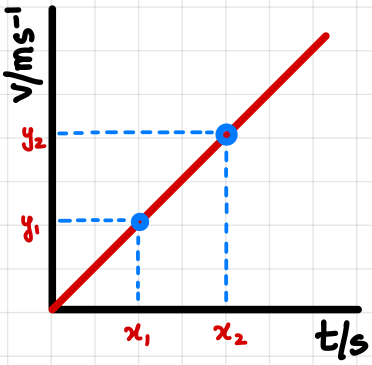


Increasing velocity



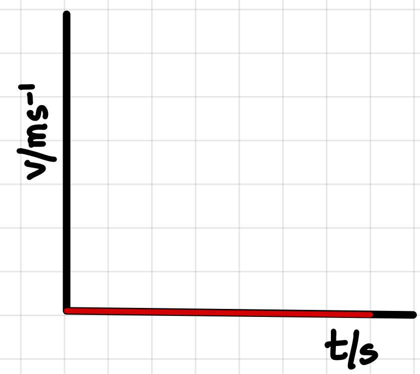
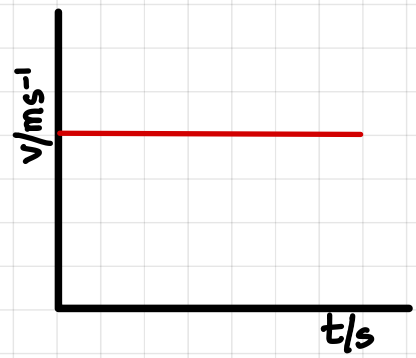
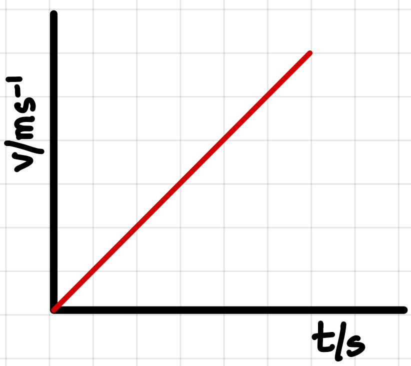
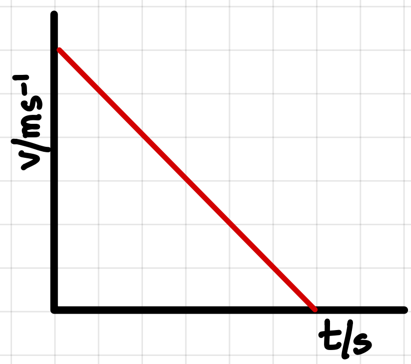
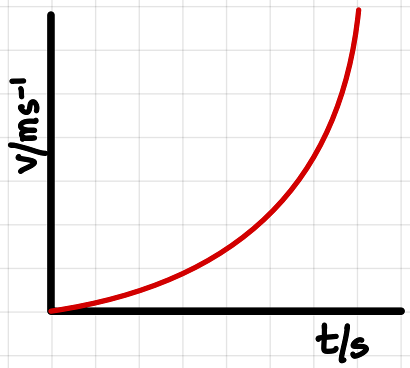
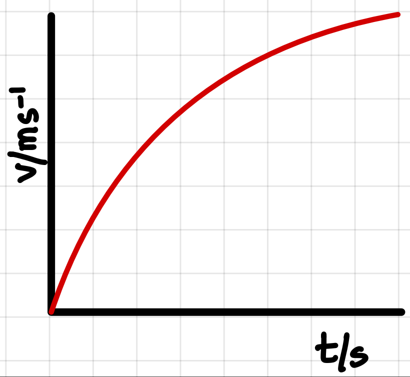
Decreasing Velocity

Velocity Time Graph:

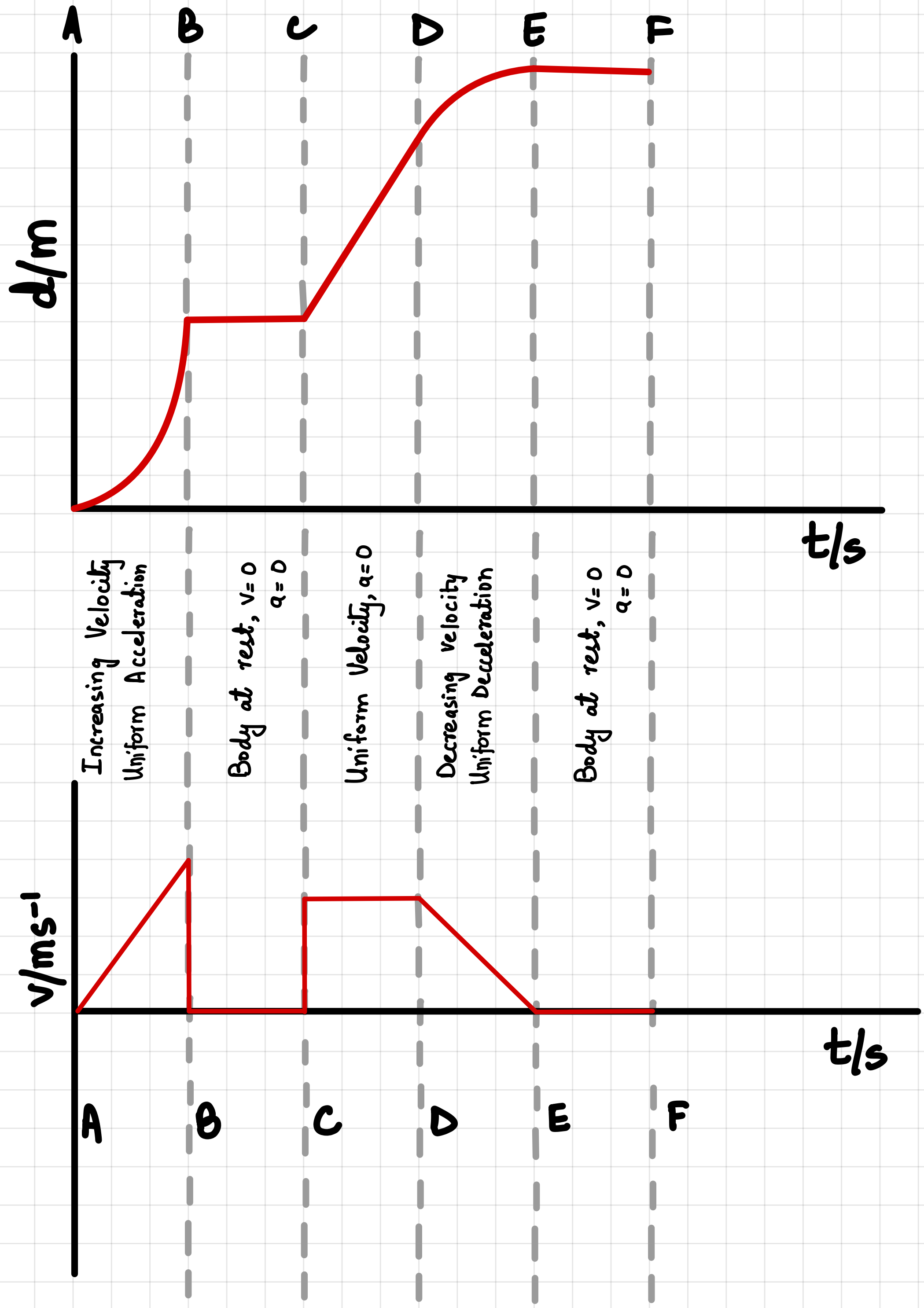


Δ Delta (Represents difference).

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\Delta v}{\Delta t} = \text{Acceleration}$$

		
Body at rest $a=0$	Uniform velocity $a=0$	Uniform Acceleration
		
Uniform Decceleration	Increasing Acceleration	Decreasing Acceleration

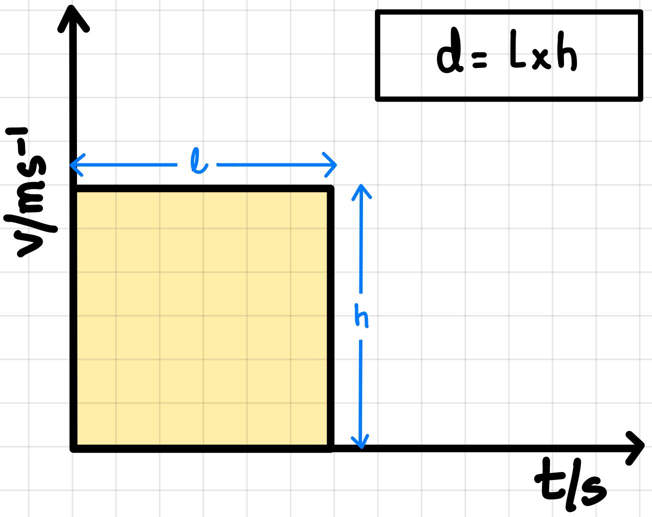
Displacement-Time → Velocity Time Conversion:



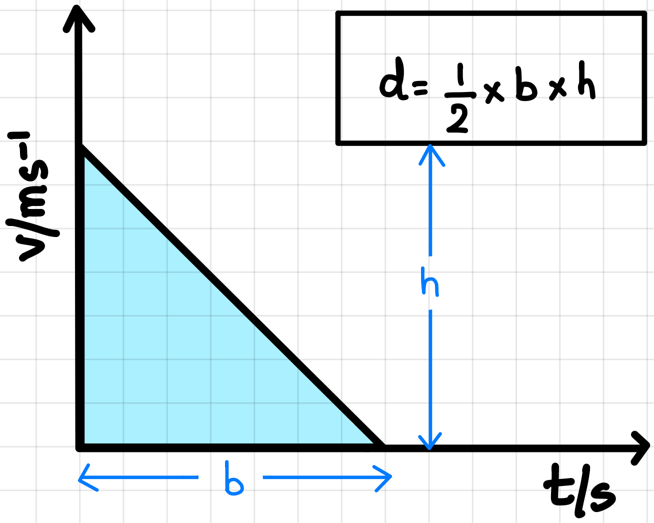
Area Under The Graph:

When velocity time graph is given & distance needs to be found.

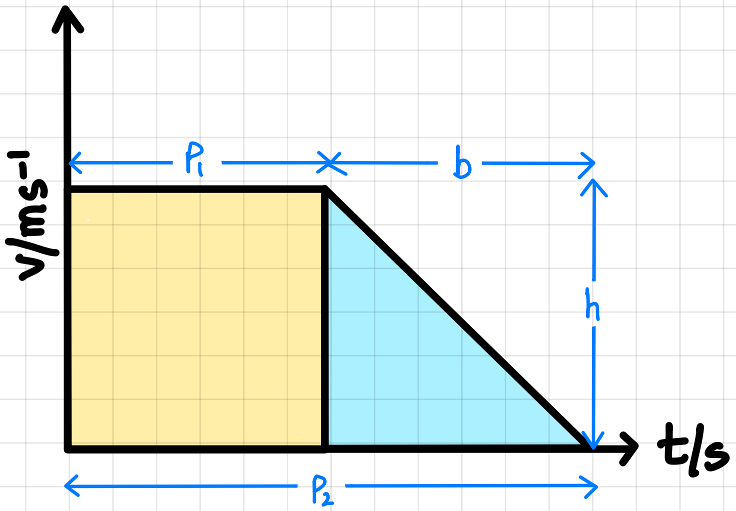
- Identify the shape of graph.
- Apply formula for area of shape to find distance.
- Shade the area whose distance needs to be found.



$d = L \times h$



$d = \frac{1}{2} \times b \times h$



$d = \frac{1}{2} \times (P_1 + P_2) \times h$

OR

$d = P_1 \times h + \frac{1}{2} \times b \times h$

Free Fall Motion:

"Irrespective to the mass of the bodies all bodies fall at the same time with same acceleration of free fall."

For Earth its $9.8\ m/s^2$.

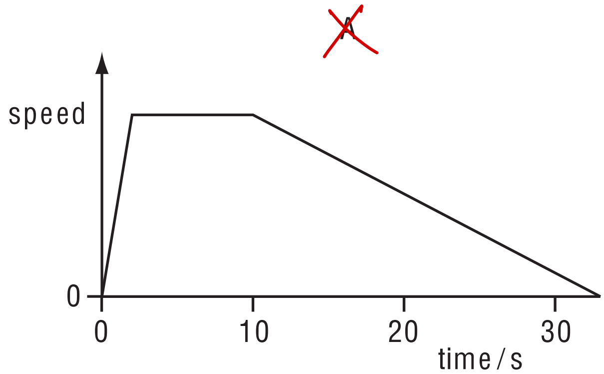
MATCH THE FOLLOWING

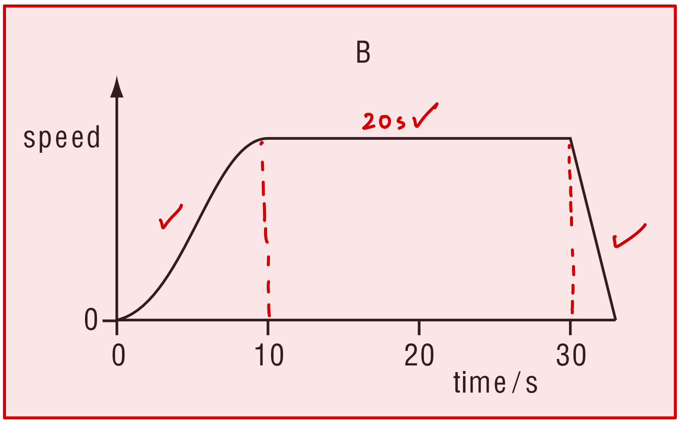
MOIZ DAWOODI
O/A LEVEL PHYSICS

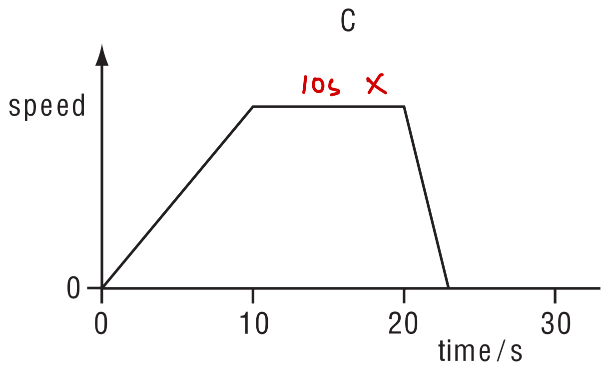
Displacement is the	covered per unit time.
Velocity is the displacement	shortest distance from starting to ending point.
Average speed is	is acceleration.
Gradient of speed-time graph	total distance covered per total time.
Gradient of distance-time graph	free fall is 9.8 m/s^2.
Value of acceleration of	is speed.
Area under the graph is	same change in displacement per unit time.
Uniform velocity is	applied in velocity-time graph & distance needs to be found.

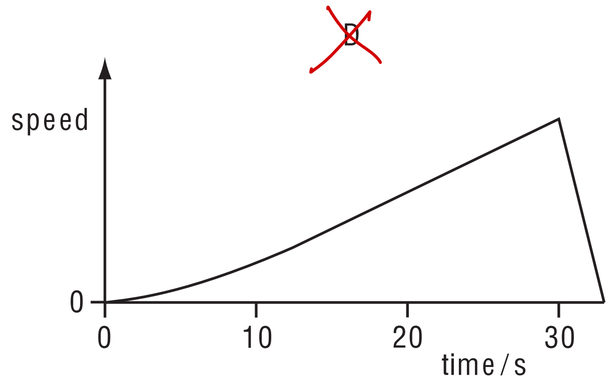
1 A car accelerates from traffic lights for 10 s. It stays at a steady speed for 20 s and then brakes to a stop in 3 s.

Which graph shows the journey?









[MJ2011/P11/Q4]

2 Which vehicle has an acceleration of $5\,m/s^2$? $a = \frac{v-u}{t}$

A a bicycle, when its speed changes from rest to 2.5 m/s in 2 s $\frac{2.5-0}{2}$

B a car, when its speed changes from rest to 15 m/s in 5 s $\frac{15-0}{5}$

C a lorry, when its speed changes from rest to 20 m/s in 15 s $\frac{20-0}{15}$

D a motorbike, when its speed changes from rest to 50 m/s in 10 s $\frac{50-0}{10}$

[MJ2011/P11/Q5]

3 The graph shows part of a journey made by a cyclist.

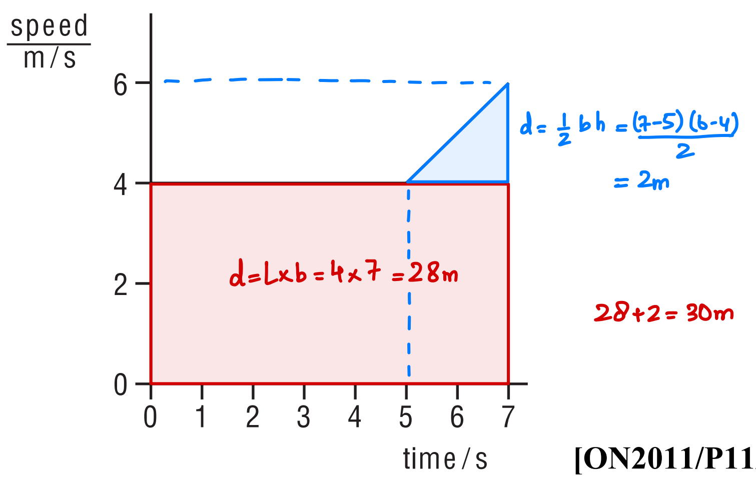
How far did the cyclist travel in 7 s?

A 28 m

B 30 m

C 32 m

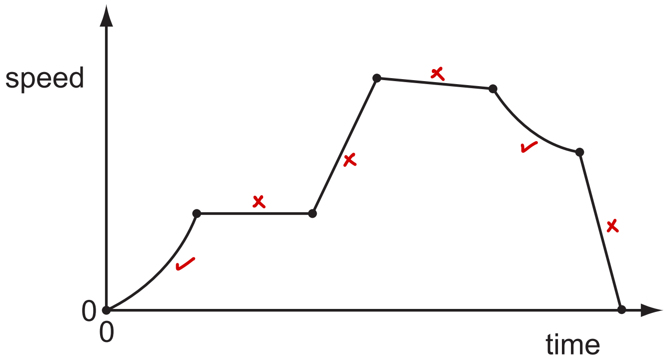
D 42 m



[ON2011/P11/Q3]

4 The speed-time graph represents the journey of a car.

The dots separate different sections of the journey. There are six different sections.



How many sections represent the car moving with non-uniform acceleration?

A 0 **B** 1 **C** 2 **D** 3 **[MJ2012/P11/Q3]**

5 A steel ball is released just below the surface of thick oil in a cylinder.

During the first few centimetres of travel, what is the acceleration of the ball?

A constant and equal to $10\,m/s^2$

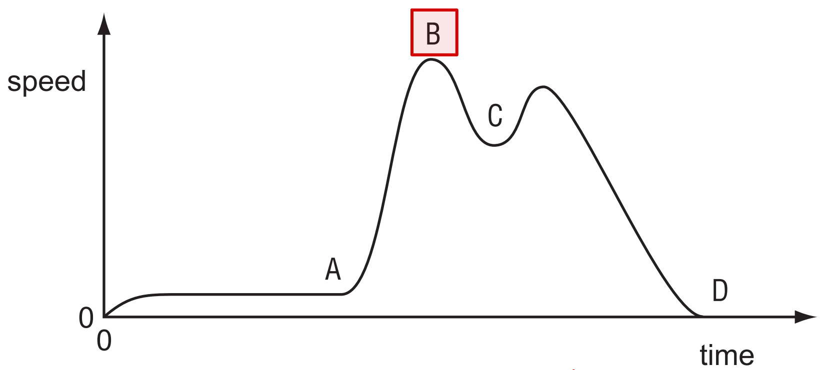
B constant but less than $10\,m/s^2$

C decreasing

D increasing **[MJ2012/P11/Q4]**

6 A cyclist travels along a hilly road without using the pedals or brakes. Air resistance and friction are negligible. The speed / time graph of the cyclist is shown.

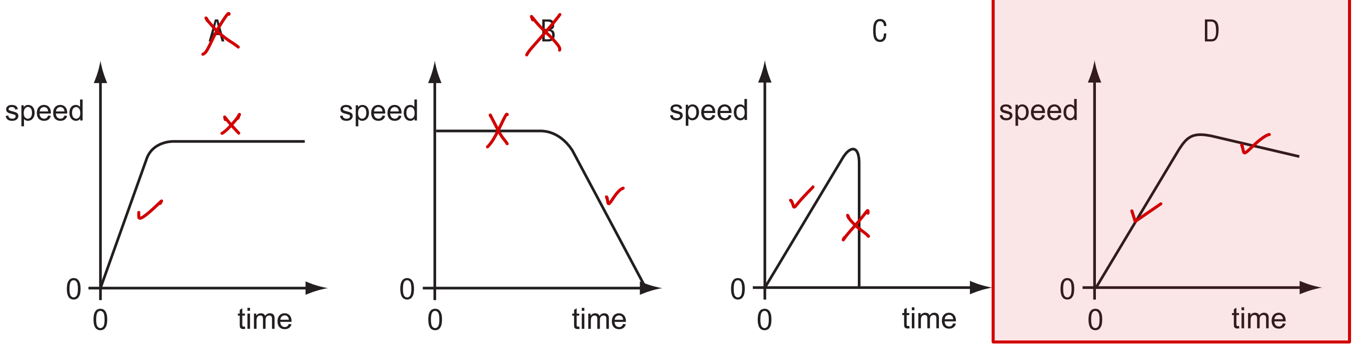
At which point did he reach the bottom of the first hill?



[MJ2012/P12/Q3]

accelerating a ↓

7 A ball starts to roll down a steep slope and then along rough horizontal ground.
Which graph best shows the **speed** of the ball?



[ON2012/P11/Q3]

8 A car is stationary at a set of traffic lights.

a=0, v=0 v↑ a↑

When the lights turn green, the car begins to move and continues to speed up until it reaches the maximum speed allowed. It continues to travel at this constant speed for the rest of the journey.

a↑ v↑ V=const a=0

What happens to the acceleration and to the velocity of the car during this journey?

A Both the acceleration and the velocity change.

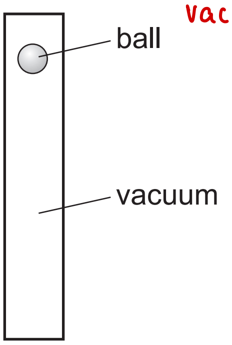
B Only the acceleration changes.

C Only the velocity changes.

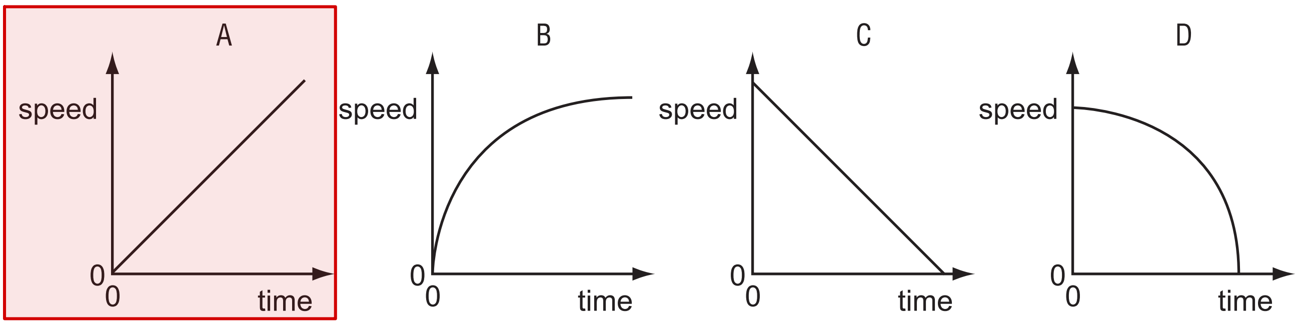
D Neither the acceleration nor the velocity changes. **[ON2012/P11/Q4]**

9 A table-tennis ball is released from the top of an evacuated tube.

Vacuum

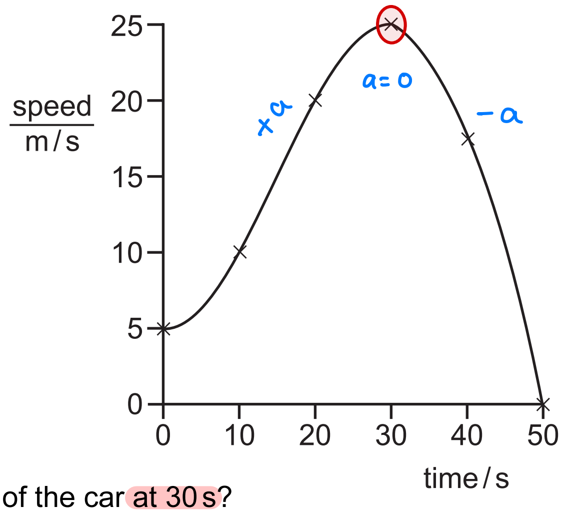


Which graph shows how the speed of the ball changes with time as it falls to the bottom of the tube?



[ON2012/P12/Q3]

10 The speed-time graph for a car is shown.



What is the acceleration of the car at 30 s?

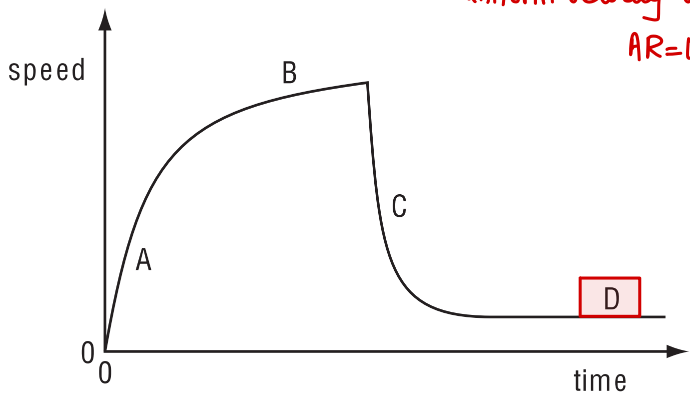
A 0 **B** $\frac{25-5}{30}$ m/s^2 **C** $\frac{25}{30}$ m/s^2 **D** $\frac{25}{50}$ m/s^2 **[ON2012/P12/Q4]**

11 The speed-time graph for a falling skydiver is shown below. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?

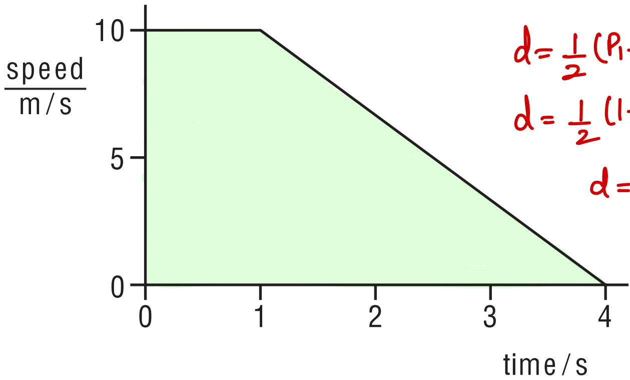
uniform velocity under free fall

AR=W



[MJ2013/P11/Q3]

12 The diagram shows the speed-time graph of the motion of a car for four seconds.



$d = \frac{1}{2}(P_1 + P_2) \times h$

$d = \frac{1}{2}(1+4) \times 10$

$d = 25\text{m}$

What is the distance travelled by the car in the four seconds?

A 15 m **B** 25 m **C** 30 m **D** 40 m [MJ2013/P12/Q4]

13 An object moves from P to Q in 10 s with uniform acceleration.

velocity at P = 5 m/s

velocity at Q = 12 m/s

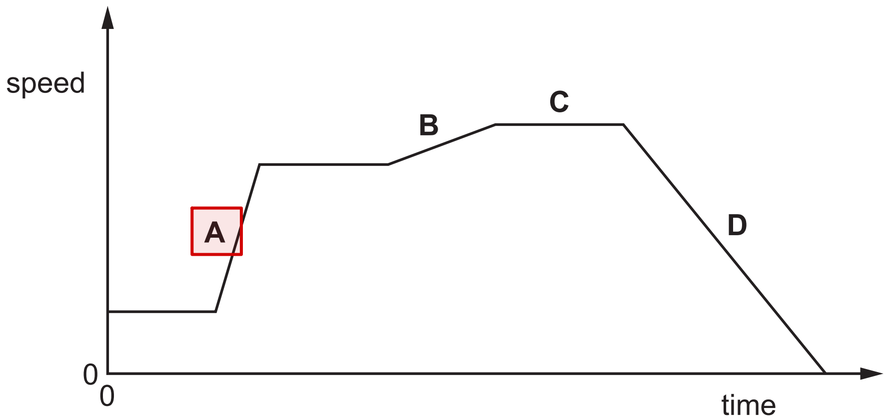
$a = \frac{v-u}{t} = \frac{12-5}{10} = \frac{7}{10}$

What is the acceleration?

A 0.5 m/s^2 **B** 0.7 m/s^2 **C** 1.2 m/s^2 **D** 1.7 m/s^2 [ON2013/P11/Q3]

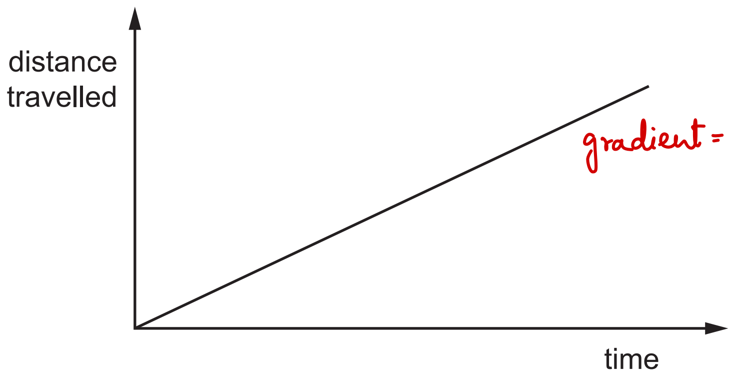
14 The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



[MJ2014/P11/Q3]

15 The distance travelled by a car is increasing uniformly as it is driven along a straight road up a hill.



gradient = velocity (constant)

Which quantity for the car is constant but not zero?

~~**A**~~ acceleration $a = 0$

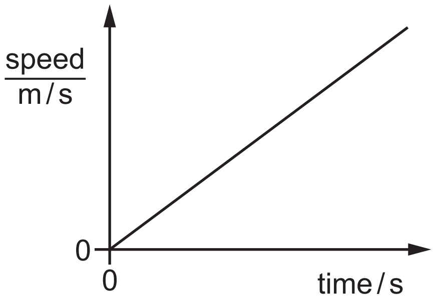
~~**B**~~ gravitational potential energy ↑

$\boxed{\textbf{C} \quad \text{kinetic energy} \quad \frac{1}{2}mv^2}$

~~**D**~~ resultant force $a = 0, \Sigma F = 0$

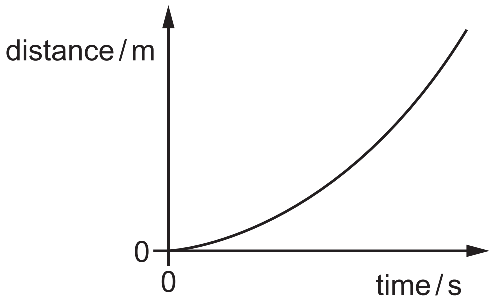
[MJ2014/P12/Q6]

16 The speed-time graph represents a short journey.

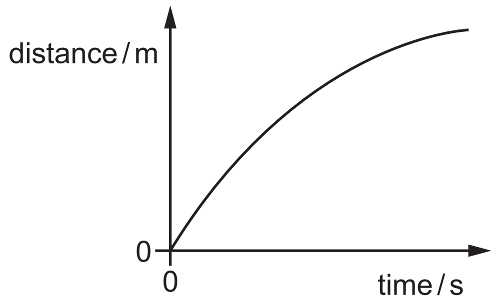


Which distance-time graph represents the same journey?

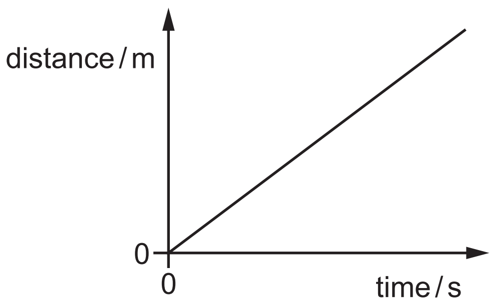
A



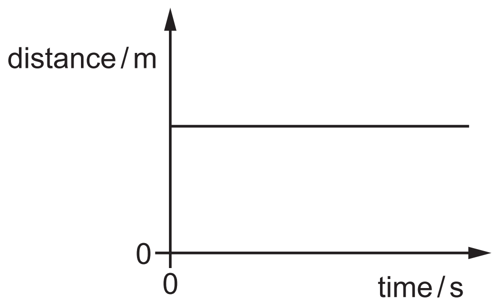
B



C



D



[ON2014/P11/Q4]

17 A student walks at a constant speed. He takes 100 s to walk 160 paces. The length of each pace is 0.80 m.

How far does the student walk in 50 s?

A 64 m **B** 80 m **C** 128 m **D** 256 m **[MJ2015/P11/Q3]**

18 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

A Both the acceleration and the velocity change.

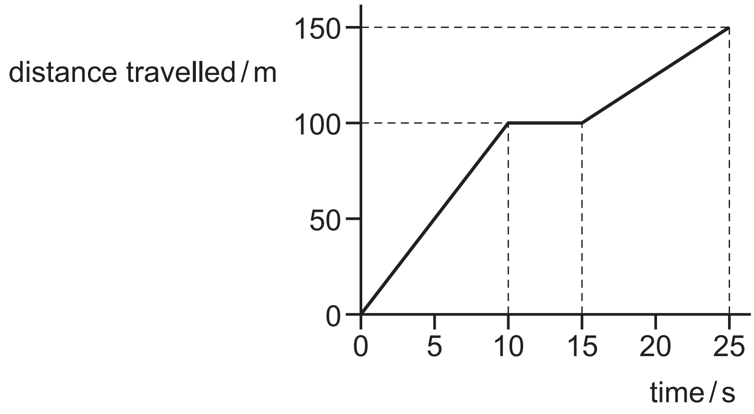
B Only the acceleration changes.

C Only the velocity changes.

D Neither the acceleration nor the velocity changes. **[MJ2015/P12/Q4]**

19 A cyclist takes a ride lasting 25 s.

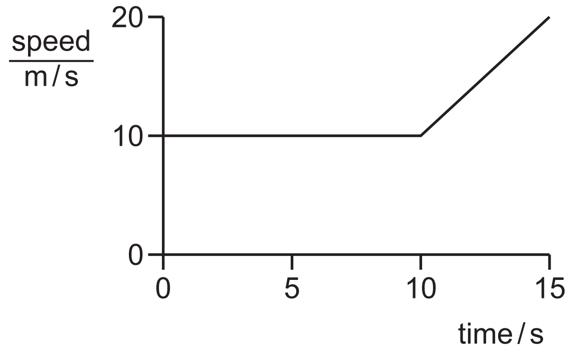
The diagram shows how her distance travelled from the starting position varies with time.



What is her average speed for the whole ride?

A 6.0 m / s **B** 7.5 m / s **C** 10.0 m / s **D** 11.0 m / s **[MJ2015/P12/Q3]**

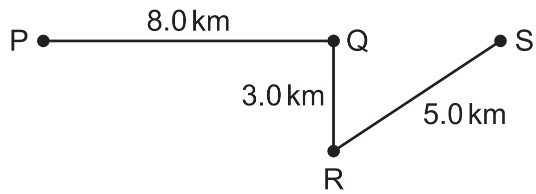
20 An object travels at a constant speed of 10 m / s for 10 s. During the next 5 s, it accelerates uniformly to 20 m / s.



What is the total distance travelled by the object?

A 150 m **B** 175 m **C** 200 m **D** 300 m **[ON2015/P11/Q3]**

21 A lorry takes 15 minutes to travel along the path PQRS.



What is the average speed of the lorry?

A 4.0 km / h **B** 22 km / h **C** 48 km / h **D** 64 km / h **[ON2015/P12/Q4]**

ANSWER KEY			
1	B	11	D
2	D	12	B
3	B	13	B
4	C	14	A
5	C	15	C
6	B	16	A
7	D	17	A
8	A	18	A
9	A	19	A
10	A	20	B
		21	D

1 Fig. 1.1 shows the speed-time graph for a car.

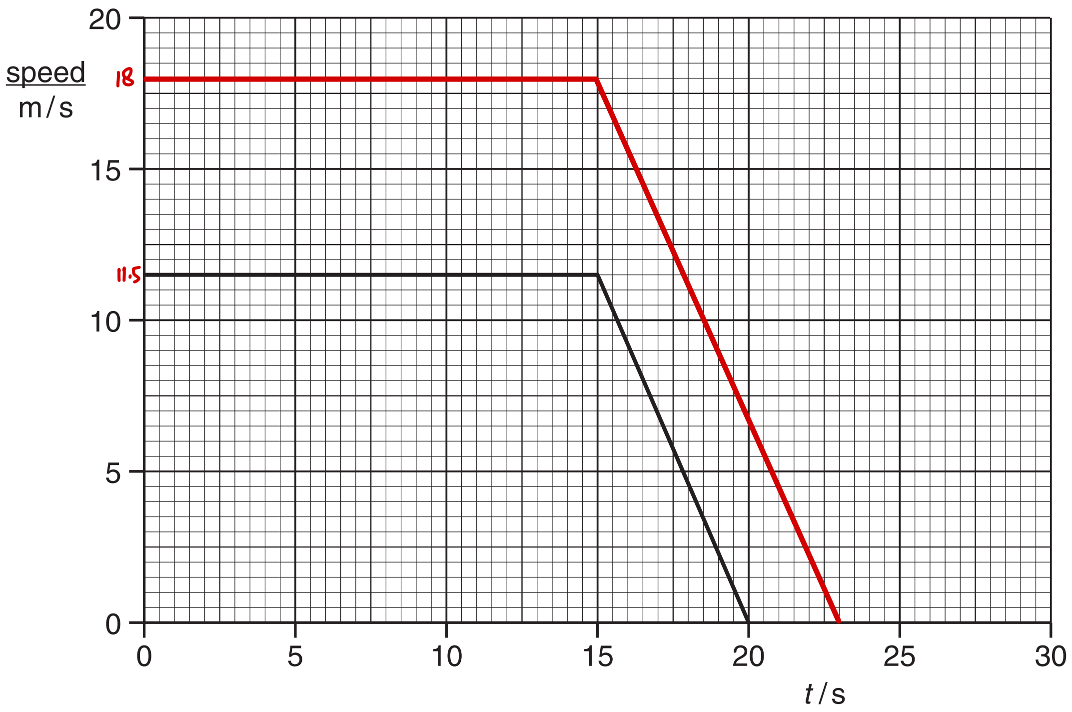


Fig. 1.1

At time t = 15 s, the brakes are applied and the car comes to rest with uniform deceleration.

(a) (i) State the speed of the car just before the brakes are applied.

speed = 11.5 $\frac{m}{s}$ [1]

(ii) Explain what is meant by *uniform deceleration*.

Same decrease in velocity in every interval of time.

[2]

(b) A lorry travels at a constant speed of 18 m/s for 15 s. At time t = 15 s, the brakes are applied and the lorry slows down with the same deceleration as the car.

(i) On Fig. 1.1, draw the speed-time graph for the lorry. [2]

(ii) Explain how your graph shows that, while braking, the lorry travels further than the car.

The area under the graph covered by lorry is greater.

[1]

[MJ2011/P21/Q1]

2 An astronaut standing on the Moon throws a stone vertically upwards. The stone leaves his hand at time t = 0. The line on Fig. 2.1 shows how the velocity v of the stone varies with time t until t = 2.0 s.

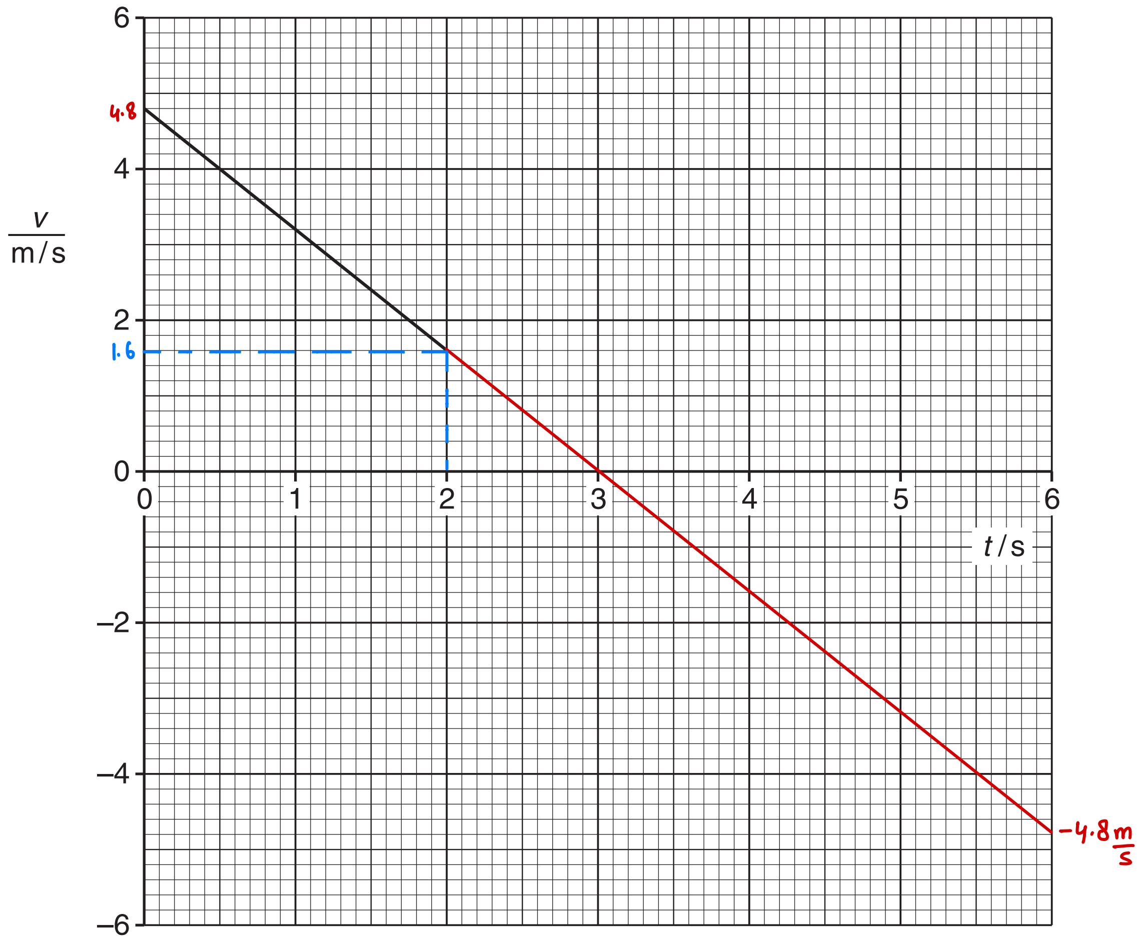


Fig. 2.1

(a) After rising, the stone falls. The astronaut catches the stone at t = 6.0 s.
There is no air resistance on the Moon.

(i) Complete Fig. 2.1 until t = 6.0 s. [1]

(ii) State the value of t when the stone is at its highest point.

3.0 s ....[1]

(b) Calculate the acceleration of the stone between t = 0 and t = 2.0 s.

$a = \frac{v-u}{t} = \frac{1.6-4.8}{2}$

acceleration = $-1.6\ \frac{m}{s^2}$ ....[2]

(c) A stone is thrown vertically upwards on the Earth with the same initial velocity.
State two ways in which the velocity-time graph for this stone differs from Fig. 1.1.

1. Acceleration will not be uniform. / Less maximum height achieved.

2. Graph will be having more gradient. ....[2]

[MJ2012/P21/Q1]

3 A ball rolls down a slope, as shown in Fig. 3.1.

The metre rule shows the position of the ball at times $t = 0$, 1.0 s, 2.0 s and 3.0 s.

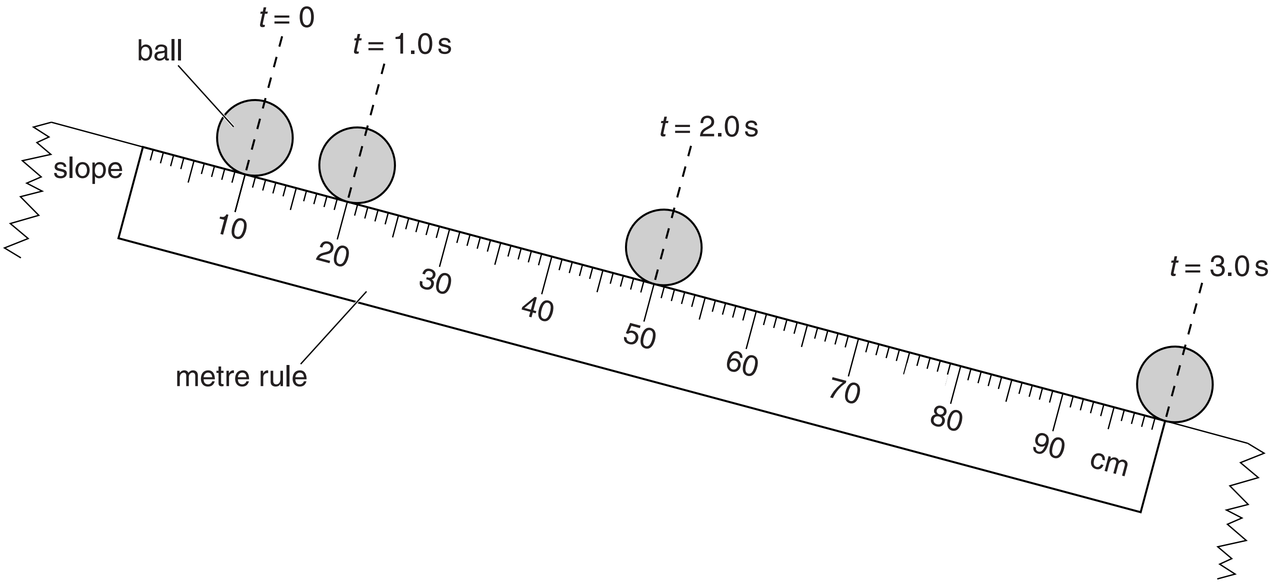


Fig. 3.1

(a) Explain how Fig. 3.1 shows that the ball is accelerating.

Distance covered per unit time is increasing, hence velocity is increasing .... [1]

(b) Calculate the average speed of the ball between $t = 1.0$ s and 3.0 s.

$$\langle v \rangle = \frac{T.d}{T.t} = \frac{80}{2} = 40 \frac{cm}{s}$$

average speed = $40 \frac{cm}{s}$ .... [2]

(c) Two of the forces that act on the ball are air resistance and weight.

State what, if anything, happens to these forces as the ball accelerates.

air resistance: Increases.

weight: Remains Same

[2]

(d) Explain why, if the slope is long enough, the ball eventually reaches a constant speed.

Forward force equals to backward force hence resultant force is zero. So acceleration is zero which causes uniform speed. [1]

[MJ2013/P22/Q1]

4 Fig. 4.1 shows a lorry accelerating in a straight line along a horizontal road.

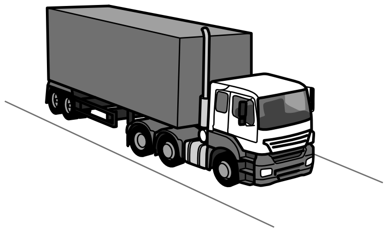


Fig. 4.1

(a) The driving force on the lorry in the forward direction is *D* and the total backward force on the lorry is *B*.

(i) State and explain whether *D* or *B* is the larger force.

D is greater than B as the lorry is accelerating.

.... [1]

(ii) Suggest one possible cause of the backward force *B*.

Air Resistance / Friction b/w road & tyres. [1]

(b) The weight of the lorry is 300 000 N.

The gravitational field strength *g* is 10 N/kg.

(i) Calculate the mass of the lorry.

$$m = \frac{W}{g} = \frac{300{,}000}{10}$$

mass = 30,000 kg [1]

(ii) The resultant force on the lorry is 15 000 N. Calculate the acceleration of the lorry.

$$\Sigma F = ma$$

$$a = \frac{\Sigma F}{m} = \frac{15000}{30{,}000}$$

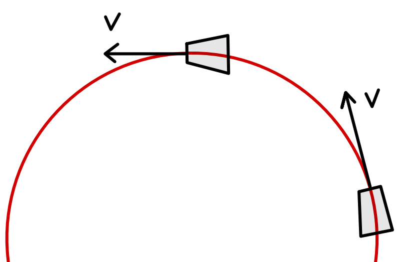
acceleration = $0.5\ m/s^2$ [2]

(c) Later, the lorry turns a corner at constant speed.

Explain why the lorry accelerates even though the speed is constant.

As car turns, velocity changes due to change in direction so acceleration is produced (which always towards the center). [1]

[MJ2014/P22/Q1]



5 Fig. 5.1 shows the distance-time graph for two cyclists A and B. They start a 500 m race together but finish the race at different times.

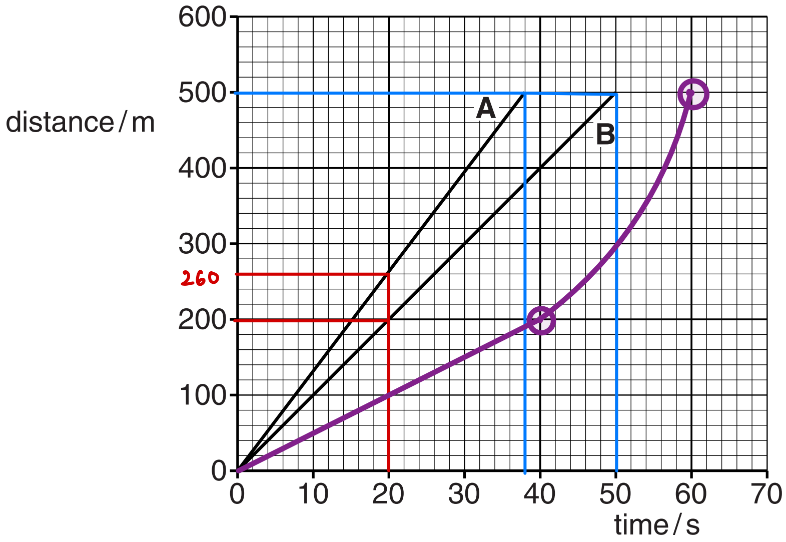


Fig. 5.1

(a) Use Fig. 5.1 to determine

(i) the distance between A and B at time $t = 20$ s,

60m ....[1]

(ii) the difference in the time taken by A and B for the race.

12 s ....[1]

(b) Cyclist C starts the race at the same time as A and B and covers the first 200 m of the race at a constant speed of 5.0 m / s. He then accelerates and finishes the race at $t = 60$ s.

(i) On Fig. 5.1, draw the distance-time graph for cyclist C. $t = \frac{d}{v} = \frac{200}{5} = 40s$ [2]

(ii) Calculate the average speed of cyclist C for the whole race.

$\langle v \rangle = \frac{T.d}{T.t} = \frac{500}{60}$

speed = $8.3\ \frac{m}{s}$ ....[2]

[MJ2015/P21/Q1]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	21

1 (a) (i) 11.5 m/s B1
(ii) decrease in speed // negative gradient C1
equal changes/decreases in speed in the same time // const. neg. grad. on v-t graph A1
(b) (i) flat line at 18 m/s from t = 0 to 15 B1
constant slope downwards parallel to initial line (by eye) B1
(ii) greater area **under** graph // higher **initial/average** speed B1 [6]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	21

1 (a) (i) straight line continues to 6 ± 0.2 s B1
(ii) 3(.0) s OR the time on Fig. 1.1 when $v = 0$ B1
(b) (a =) ($v – u$)/t in any form numerical or algebraic C1
(–)1.6 m/s^2 A1
(c) any TWO lines:
(at first) graph steeper/higher acceleration/deceleration
caught sooner/shorter time to maximum
graph curves (due to air resistance) B2 [6]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2013	5054	22

1 (a) travels further in each second / in same time / between images B1

(b) (s=) d/t in any form algebraic or numerical C1
40 cm/s; 0.4(0) m/s A1

(c) air resistance increases B1
weight constant B1

(d) forces balance /cancel B1 [6]
or no resultant/net force
or resultant of any two forces equal and opposite to third

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2014	5054	22

1 (a) (i) *D* **and either** lorry accelerates (forward) **or** resultant force is forward [B1]
(ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]
(b) (i) 30 000 kg [B1]
(ii) (a=) F/m algebraic in any form or numerical [C1]
0.5(0) m/s^2 [A1]
(c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21

1 (a) (i) 60 m B1
(ii) 12 s B1
(b) (i) straight line from origin to 200 m at 40 s B1
any line straight or curved from (40,200) to (60,500) B1
(ii) $s = d/t$ or 500/60 C1
8.3 m/s A1